

New Media Short Questions

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1. New Media

Definition:

New Media refers to digital and internet-based communication technologies that allow people to create, share, and interact with content in real-time. It includes social media, websites, online news, and digital videos.

Origin:

- The concept of **New Media** started in the **1950s-60s** with the development of computers and early digital technology.
- The term "New Media" became popular in the **1990s** with the rise of the internet and digital communication.

History:

- **1950s-60s:** Early computer networks were developed.
- **1980s:** Personal computers became more common, and digital media started growing.
- **1990s:** The internet and email changed the way people communicated.
- **2000s-Present:** Social media, streaming platforms, and digital news transformed media consumption.

Key Features of New Media:

1. **Digital:** Information is stored and shared electronically.
2. **Interactive:** Users can engage by liking, commenting, and sharing content.
3. **Real-time Communication:** Instant messaging and live streaming are possible.
4. **User-generated Content:** Anyone can create and publish content (e.g., blogs, vlogs, podcasts).
5. **Multi-platform:** Accessible on smartphones, computers, and smart devices.

Examples of New Media:

- Social media (Facebook, Instagram, Twitter).
- Online news websites (BBC, CNN, The Hindu).
- Video streaming platforms (YouTube, Netflix).
- Digital advertising and e-books.

2. Big Data

Definition:

Big Data refers to a huge amount of information that is collected from different sources and processed using advanced technology. It is too large and complex for traditional computers to handle.

Origin:

- The concept of **Big Data** started in the **1990s** when scientists and researchers began studying large datasets.
- The term "**Big Data**" became widely used in the early **2000s** as companies started using data for business growth.

History:

- **1960s-70s:** Businesses and governments began storing data in computers.
- **1980s-90s:** The rise of databases and data warehouses helped manage large amounts of information.
- **2000s:** The internet, social media, and e-commerce led to an explosion of data.
- **2010s-Present:** Technologies like cloud computing, artificial intelligence (AI), and machine learning help analyze and use Big Data efficiently.

Key Features (3Vs of Big Data):

1. **Volume:** Huge amounts of data generated every second.
2. **Velocity:** Data is created and processed at high speed.
3. **Variety:** Data comes in different formats (text, images, videos, numbers, etc.).

Examples of Big Data:

- Google searches and YouTube recommendations.
- Social media posts on Facebook, Instagram, Twitter.
- Online shopping data from Amazon, Flipkart.
- Health records and sensor data from smartwatches.

3. Dark Net

Definition:

The **Dark Net** is a hidden part of the internet that is not accessible through normal web browsers like Google Chrome. It requires special software, such as **Tor (The Onion Router)**, to access. It is used for both legal and illegal activities, allowing users to stay anonymous.

Origin:

- The concept of the **Dark Net** started in the **1970s** when the **ARPANET** (early internet) allowed private communication between computers.
- The term "Dark Net" became more popular in the early **2000s** with the development of **Tor** by the U.S. Naval Research Lab for secure communication.

History:

- **1970s:** ARPANET researchers experimented with private and anonymous communication.
- **1990s:** Development of encryption tools for secure online interactions.
- **2000s:** Tor and other networks made anonymous browsing more accessible.
- **2010s-Present:** The Dark Net grew, used by whistleblowers, journalists, activists, but also criminals.

Key Features:

1. **Anonymity:** Users can browse and communicate without revealing their identity.
2. **Not Indexed:** Websites on the Dark Net do not appear in Google search results.
3. **Special Access Required:** Only accessible through tools like **Tor** or **I2P** (Invisible Internet Project).
4. **Used for Various Purposes:** Includes legal uses (privacy, journalism) and illegal uses (hacking, drug trade).

Examples of Dark Net Use:

Legal Uses: Secure communication for journalists and activists, bypassing censorship.
Illegal Uses: Cybercrime, drug trafficking, hacking services.

Notable Cases in India:

1. **2019 (Delhi):**

- NCB arrested a Hyderabad-based engineer for **buying LSD using Bitcoin** via the Darknet.

2. **2020 (Bengaluru):**

- A drug racket was busted where techies were **selling MDMA using cryptocurrency**.

3. **2021 (Mumbai):**

- ED traced **cryptocurrency fraud** linked to Darknet transactions involving hacking services and fake passports.

4. **2022 (Maharashtra & Delhi):**

- CBI arrested multiple suspects in an **international child pornography ring** operating via Darknet forums.

5. **2023 (Multiple Cities):**

- Cybercriminals were selling **hacking services, fake Aadhaar cards, and leaked banking details**.

4. Dark Web

Definition:

The **Dark Web** is a hidden part of the internet that cannot be accessed through normal web browsers like Google Chrome or Bing. It requires special software like **Tor (The Onion Router)** to access and is known for providing anonymity to users. The Dark Web contains both legal and illegal content.

Origin:

- The concept of the **Dark Web** started in the **1990s** when researchers developed encrypted networks for secure communication.
- The U.S. Naval Research Laboratory created **Tor** in the early **2000s** to help protect online privacy.

History:

- **1990s:** Early experiments with encrypted and private networks.
- **2002:** Tor was developed to allow secure and anonymous browsing.
- **2009-Present:** The Dark Web became widely known, used by journalists, activists, and cybercriminals.

Key Features:

1. **Anonymity:** Users can browse without revealing their identity.
2. **Not Indexed:** Dark Web websites do not appear on Google or other search engines.
3. **Special Access Required:** Can only be accessed using software like **Tor** or **I2P**.
4. **Wide Range of Content:** Includes legal uses (private communication, research) and illegal activities (black markets, hacking forums).

Examples of Dark Web Use:

Legal Uses: Private messaging, whistleblowing, bypassing internet censorship.

Illegal Uses: Black markets, cybercrime, identity theft

Difference Between Darknet and Dark Web

Though often used interchangeably, **Darknet** and **Dark Web** have different meanings.

Feature	Darknet	Dark Web
Definition	A network of encrypted and anonymous online connections.	A part of the internet that exists on the darknet, containing hidden websites.
Access	Requires special software like Tor , I2P , Freenet .	Accessed through darknet networks using special tools like Tor browser .
Purpose	Designed for private, anonymous, and censorship-free communication .	Hosts hidden websites for legal and illegal activities.
Usage	Used for anonymous communication, secure file sharing, and bypassing censorship.	Used for secretive websites like illegal marketplaces, hacking forums, whistleblowing platforms . Some activities are legal (e.g., journalism), but many are illegal (e.g., drug trafficking, cybercrime).
Legality	Not illegal in itself; used for privacy and free speech.	
Example	Tor Network, I2P, Silk Road, Freenet (provides drug market),	

Feature	Darknet	Dark Web
	infrastructure (hidden sites).	for Hidden Wiki (directory of dark web sites) .

5. E-Learning

Definition:

E-Learning (Electronic Learning) is a method of education where people use digital devices and the internet to study from anywhere. It includes online courses, video lectures, virtual classrooms, and educational apps.

Origin:

- The concept of **E-Learning** started in the **1960s** with early computer-based training programs.
- The term "E-Learning" became popular in the **1990s** when the internet was used for education.

History:

- **1960s-70s:** Early computer-based learning systems were developed.
- **1980s:** CD-ROMs were used for interactive learning.
- **1990s:** The internet introduced online education and virtual classrooms.
- **2000s-Present:** Mobile learning, AI-powered platforms, and video lectures became popular.

Key Features of E-Learning:

1. **Digital Learning:** Education is available through computers, mobile phones, and tablets.
2. **Flexibility:** Students can learn anytime, anywhere.
3. **Interactive Content:** Videos, quizzes, and live sessions make learning engaging.
4. **Self-paced Learning:** Students can learn at their own speed.
5. **Cost-effective:** Reduces the need for physical classrooms and printed materials.

Government E-Learning Resources in India:

1. **SWAYAM (swayam.gov.in):** Free online courses from Indian universities.
2. **DIKSHA (diksha.gov.in):** Digital learning for school students and teachers.

3. **NPTEL (nptel.ac.in):** Online courses for engineering and science students.
4. **E-pg pathshala (epathshala.nic.in):** Digital books and resources for students.
5. **National Digital Library (ndl.iitkgp.ac.in):** Free access to books, research papers, and learning materials.
6. **eVidya (PM eVidya):** TV and online education for school students.
7. **Sakshat (saksat.ac.in)**
8. **Swayam Prabha**
9. **E-Gyankosh**
10. **OER Commons**

4. **Wide Variety of Products:** Includes electronics, clothes, groceries, and more.
5. **Customer Reviews & Ratings:** Buyers can check reviews before purchasing.
6. **Secure Transactions:** Uses encryption and cybersecurity to protect payments.

Examples of Web Commerce Platforms:

1. Amazon

- **Owner/Founder:** Jeff Bezos
- **Established Year:** 1994
- **Present CEO:** Andy Jassy
- **Tagline:** "Work Hard. Have Fun. Make History."
- **Purpose:** Online retail marketplace for a wide range of products including electronics, books, clothing, and household items.

2. Flipkart

- **Owner/Founder:** Sachin Bansal & Binny Bansal
- **Established Year:** 2007
- **Present CEO:** Kalyan Krishnamurthy
- **Tagline:** "Ab Har Wish Hogi Poori"
- **Purpose:** Indian e-commerce platform for electronics, fashion, groceries, and lifestyle products.

3. eBay

- **Owner/Founder:** Pierre Omidyar
- **Established Year:** 1995
- **Present CEO:** Jamie Iannone
- **Tagline:** "Buy it. Sell it. Love it."
- **Purpose:** Online auction and shopping website for buying and selling both new and used products.

4. Alibaba

- **Owner/Founder:** Jack Ma
- **Established Year:** 1999
- **Present CEO:** Eddie Wu
- **Tagline:** "To make it easy to do business anywhere."
- **Purpose:** B2B and B2C e-commerce marketplace for wholesale and retail trade, mainly for Chinese and global businesses.

5. Snapdeal

- **Owner/Founder:** Kunal Bahl & Rohit Bansal
- **Established Year:** 2010
- **Present CEO:** Himanshu Chakrawarti

6. Web Commerce

Definition:

Web Commerce (also called **E-Commerce**) means buying and selling products or services online using websites, apps, or digital platforms. It includes online shopping, digital payments, and online business transactions.

Origin:

- The concept of **Web Commerce** started in the **1970s** with **electronic data interchange (EDI)**, which allowed businesses to exchange documents electronically.
- The first real online sale happened in **1994**, when a man bought a CD from a website using a credit card.

History:

- **1970s-80s:** Businesses started using electronic data transfer for transactions.
- **1990s:** The internet became popular, and companies like **Amazon (1994)** and **eBay (1995)** started online selling.
- **2000s:** Growth of secure online payments like **PayPal** and mobile shopping.
- **2010s-Present:** Rise of digital wallets (Google Pay, UPI), mobile apps, and AI-powered shopping experiences.

Key Features of Web Commerce:

1. **Online Transactions:** Buying and selling take place through websites and apps.
2. **Digital Payments:** Supports credit cards, debit cards, UPI, net banking, and e-wallets.
3. **24/7 Availability:** Customers can shop anytime, from anywhere.

- **Tagline:** "Brand Waali Quality, Bazaar Waali Deal"
- **Purpose:** Indian online shopping platform for fashion, electronics, and household products at affordable prices.

6. Myntra

- **Owner/Founder:** Mukesh Bansal, Ashutosh Lawania & Vineet Saxena
- **Established Year:** 2007
- **Present CEO:** Nandita Sinha
- **Tagline:** "India's Fashion Expert"
- **Purpose:** Online fashion and lifestyle shopping platform, offering clothing, accessories, and footwear.

7. Shopify

- **Owner/Founder:** Tobias Lütke, Daniel Weinand & Scott Lake
- **Established Year:** 2006
- **Present CEO:** Harley Finkelstein
- **Tagline:** "The future of commerce is here."
- **Purpose:** E-commerce platform enabling businesses to create online stores and manage digital sales.

8. Paytm Mall

- **Owner/Founder:** Vijay Shekhar Sharma
- **Established Year:** 2017
- **Present CEO:** Varun Sridhar
- **Tagline:** "Ab Shopping Hogi Budget Friendly"
- **Purpose:** Indian e-commerce platform for electronics, fashion, and daily essentials with cashback offers.

9. Nykaa

- **Owner/Founder:** Falguni Nayar
- **Established Year:** 2012
- **Present CEO:** Falguni Nayar
- **Tagline:** "Your Beauty, Our Passion"
- **Purpose:** Online platform specializing in beauty, skincare, cosmetics, and personal care products.

10. Meesho

- **Owner/Founder:** Vidit Aatrey & Sanjeev Barnwal
- **Established Year:** 2015
- **Present CEO:** Vidit Aatrey
- **Tagline:** "Not Just a Marketplace, a Revolution"

- **Purpose:** Social commerce platform enabling individuals and small businesses to resell products through WhatsApp and Facebook.

11. Reliance JioMart

- **Owner/Founder:** Mukesh Ambani (Reliance Industries)
- **Established Year:** 2020
- **Present CEO:** Mukesh Ambani
- **Tagline:** "Desh Ki Nayi Dukaan"
- **Purpose:** Online grocery and daily essentials shopping platform in India, integrated with WhatsApp for easy ordering.

12. BigBasket

- **Owner/Founder:** Hari Menon, Vipul Parekh, Abhinay Choudhari, VS Sudhakar, and VS Ramesh
- **Established Year:** 2011
- **Present CEO:** Hari Menon
- **Tagline:** "India's Largest Online Supermarket"
- **Purpose:** Online grocery delivery service offering fresh vegetables, fruits, and household essentials.

13. Zomato

- **Owner/Founder:** Deepinder Goyal & Pankaj Chaddah
- **Established Year:** 2008
- **Present CEO:** Deepinder Goyal
- **Tagline:** "Never Have a Bad Meal"
- **Purpose:** Online food ordering, restaurant discovery, and delivery service in multiple countries.

14. Swiggy

- **Owner/Founder:** Sriharsha Majety, Nandan Reddy, and Rahul Jaimini
- **Established Year:** 2014
- **Present CEO:** Sriharsha Majety
- **Tagline:** "Swiggy It"
- **Purpose:** Online food delivery and grocery delivery platform with on-demand service.

15. Dunzo

- **Owner/Founder:** Kabeer Biswas, Dalvir Suri, Ankur Aggarwal, Mukund Jha
- **Established Year:** 2015
- **Present CEO:** Kabeer Biswas
- **Tagline:** "Making India More Time-Rich"
- **Purpose:** Hyperlocal delivery service for groceries, medicines, and essential goods.

16. Blinkit (Formerly Grofers)

- **Owner/Founder:** Albinder Dhindsa & Saurabh Kumar
- **Established Year:** 2013
- **Present CEO:** Albinder Dhindsa
- **Tagline:** "India's Quick Commerce Platform"
- **Purpose:** 10-minute grocery and essentials delivery service.

17. FirstCry

- **Owner/Founder:** Supam Maheshwari & Amitava Saha
- **Established Year:** 2010
- **Present CEO:** Supam Maheshwari
- **Tagline:** "Big Store for Little Ones"
- **Purpose:** E-commerce platform specializing in baby and kids' products.

18. Pepperfry

- **Owner/Founder:** Ambareesh Murty & Ashish Shah
- **Established Year:** 2011
- **Present CEO:** Ashish Shah
- **Tagline:** "Happy Furniture to You"
- **Purpose:** Online furniture and home decor shopping platform.

19. Lenskart

- **Owner/Founder:** Peyush Bansal, Amit Chaudhary, Sumeet Kapahi
- **Established Year:** 2010
- **Present CEO:** Peyush Bansal
- **Tagline:** "The Joy of Perfect Vision"
- **Purpose:** Online eyewear store for glasses, contact lenses, and sunglasses.

20. UrbanClap (Now Urban Company)

- **Owner/Founder:** Abhiraj Bhal, Raghav Chandra & Varun Khaitan
- **Established Year:** 2014

- **Present CEO:** Abhiraj Bhal
- **Tagline:** "Your Home Services Expert"
- **Purpose:** Online platform for booking home services like beauty, repair, and cleaning.

21. 1mg (Now Tata 1mg)

- **Owner/Founder:** Prashant Tandon, Gaurav Agarwal, and Vikas Chauhan
- **Established Year:** 2015
- **Present CEO:** Prashant Tandon
- **Tagline:** "Health Comes First"
- **Purpose:** Online pharmacy and healthcare service platform.

22. PharmEasy

- **Owner/Founder:** Dharmil Sheth, Dhaval Shah, and Mikhil Innani
- **Established Year:** 2015
- **Present CEO:** Siddharth Shah
- **Tagline:** "Take It Easy with PharmEasy"
- **Purpose:** Online medicine ordering and healthcare service provider.

23. Netmeds

- **Owner/Founder:** Pradeep Dadha
- **Established Year:** 2010
- **Present CEO:** Vishal Duggal
- **Tagline:** "India Ki Pharmacy"
- **Purpose:** Online pharmacy for medicines, wellness products, and health consultations.

24. OLX

- **Owner/Founder:** Fabrice Grinda & Alec Oxenford
- **Established Year:** 2006
- **Present CEO:** Gautam Thakar
- **Tagline:** "Bech De!"
- **Purpose:** Online classified marketplace for buying and selling used goods.

25. Quikr

- **Owner/Founder:** Pranay Chulet & Jiby Thomas
- **Established Year:** 2008
- **Present CEO:** Pranay Chulet
- **Tagline:** "Aasan Hai Badalna"
- **Purpose:** Online classified ads platform for real estate, jobs, and second-hand products.

7. Multimedia

Definition:

Multimedia means using a combination of different forms of content such as **text, images, audio, video, animation, and interactive elements** to present information. It is used in education, entertainment, business, and communication.

Origin:

- The concept of **Multimedia** started in the **1960s**, when computers began supporting text and images together.
- The term "Multimedia" became popular in the **1990s** with the rise of digital technology and the internet.

History:

- **1960s:** Early computers displayed text and images together.
- **1970s-80s:** CDs and early computer programs started including sound and animation.
- **1990s:** The internet and software like PowerPoint made multimedia widely accessible.
- **2000s-Present:** Smartphones, social media, virtual reality (VR), and AI-powered multimedia advanced digital experiences.

Key Features of Multimedia:

1. **Multiple Content Types:** Uses text, images, audio, video, and animation.
2. **Interactivity:** Users can engage with content (e.g., clicking, dragging, playing videos).
3. **Digital and Electronic Format:** Found on websites, apps, presentations, and games.
4. **Wide Applications:** Used in education, entertainment, business, and marketing.
5. **Improves Communication:** Makes learning and information sharing more engaging..

Examples of Multimedia Platforms

1. YouTube

- **Owner/Founder:** Steve Chen, Chad Hurley, and Jawed Karim (Now owned by Google)
- **Established Year:** 2005
- **Present CEO:** Neal Mohan
- **Tagline:** "Broadcast Yourself"
- **Purpose:** Video-sharing platform where users can upload, watch, and share videos, including

entertainment, education, and vlogging.

2. Netflix

- **Owner/Founder:** Reed Hastings and Marc Randolph
- **Established Year:** 1997
- **Present CEO:** Ted Sarandos and Greg Peters
- **Tagline:** "See What's Next"
- **Purpose:** Streaming service for movies, TV shows, and original content.

3. Spotify

- **Owner/Founder:** Daniel Ek and Martin Lorentzon
- **Established Year:** 2006
- **Present CEO:** Daniel Ek
- **Tagline:** "Music for Everyone"
- **Purpose:** Digital music streaming service offering songs, podcasts, and audiobooks.

4. Amazon Prime Video

- **Owner/Founder:** Amazon (Jeff Bezos)
- **Established Year:** 2006
- **Present CEO:** Mike Hopkins
- **Tagline:** "Your Favorite Movies & TV Shows, Anytime, Anywhere"
- **Purpose:** Video streaming service with original and licensed content.

5. Disney+ (Disney Plus)

- **Owner/Founder:** The Walt Disney Company
- **Established Year:** 2019
- **Present CEO:** Bob Iger
- **Tagline:** "The Best Stories in the World, All in One Place"
- **Purpose:** Streaming platform for Disney, Marvel, Star Wars, Pixar, and National Geographic content.

6. Hotstar (Now Disney+ Hotstar)

- **Owner/Founder:** Star India (Now owned by Disney)
- **Established Year:** 2015
- **Present CEO:** Sajith Sivanandan
- **Tagline:** "Go Solo or Go Big"
- **Purpose:** Online streaming platform for live sports, movies, and TV shows.

7. SonyLIV

- **Owner/Founder:** Sony Pictures Networks India
- **Established Year:** 2013
- **Present CEO:** NP Singh

- **Tagline:** "We LIV to Entertain"
- **Purpose:** Streaming platform for movies, TV shows, and sports content.

8. Vimeo

- **Owner/Founder:** Jake Lodwick and Zach Klein
- **Established Year:** 2004
- **Present CEO:** Adam Gross
- **Tagline:** "Powering Amazing Videos"
- **Purpose:** Video-sharing platform focused on professional and high-quality video content.

9. Twitch

- **Owner/Founder:** Justin Kan and Emmett Shear (Now owned by Amazon)
- **Established Year:** 2011
- **Present CEO:** Dan Clancy
- **Tagline:** "You're Already One of Us"
- **Purpose:** Live streaming platform mainly for gaming, e-sports, and creative content.

10. Apple Music

- **Owner/Founder:** Apple Inc.
- **Established Year:** 2015
- **Present CEO:** Tim Cook
- **Tagline:** "Listen Now"
- **Purpose:** Music streaming service with a vast library of songs, albums, and exclusive releases.

11. SoundCloud

- **Owner/Founder:** Alexander Ljung and Eric Wahlforss
- **Established Year:** 2007
- **Present CEO:** Eliah Seton
- **Tagline:** "Hear the World's Sounds"
- **Purpose:** Platform for independent artists to upload, share, and promote their music.

12. JioSaavn

- **Owner/Founder:** Reliance Jio and Saavn
- **Established Year:** 2007
- **Present CEO:** Rishi Malhotra
- **Tagline:** "Let the Music Play"
- **Purpose:** Indian music streaming service offering Bollywood, regional, and international songs.

13. Gaana

- **Owner/Founder:** Times Internet
- **Established Year:** 2010
- **Present CEO:** Sandeep Lodha
- **Tagline:** "Bas Bajna Chahiye Gaana"
- **Purpose:** Music streaming service offering Indian and international songs.

14. Wynk Music

- **Owner/Founder:** Bharti Airtel
- **Established Year:** 2014
- **Present CEO:** Gopal Vittal
- **Tagline:** "Your Music. Your Mood."
- **Purpose:** Music streaming service with a vast collection of songs and playlists.

15. MX Player

- **Owner/Founder:** J2 Interactive (Now owned by Times Internet)
- **Established Year:** 2011
- **Present CEO:** Karan Bedi
- **Tagline:** "Everytainment"
- **Purpose:** Video streaming platform offering free movies, web series, and TV shows.

16. ZEE5

- **Owner/Founder:** Zee Entertainment Enterprises
- **Established Year:** 2018
- **Present CEO:** Punit Goenka
- **Tagline:** "Dil Se Desi"
- **Purpose:** Indian streaming platform offering TV shows, movies, and regional content.

17. ALTBalaji

- **Owner/Founder:** Balaji Telefilms
- **Established Year:** 2017
- **Present CEO:** Nachiket Pantvaitya
- **Tagline:** "Hai Toh Bahut Hai"
- **Purpose:** Subscription-based Indian streaming platform focused on original web series and shows.

18. Eros Now

- **Owner/Founder:** Eros International
- **Established Year:** 2012

- **Present CEO:** Rishika Lulla Singh
- **Tagline:** "Now, That's Entertainment!"
- **Purpose:** Indian digital streaming platform with movies, originals, and TV content.

19. Voot

- **Owner/Founder:** Viacom18
- **Established Year:** 2016
- **Present CEO:** Gourav Rakshit
- **Tagline:** "Made for Stories"
- **Purpose:** Indian streaming service with TV shows, web series, and movies.

20. Resso

- **Owner/Founder:** ByteDance (Owner of TikTok)
- **Established Year:** 2020
- **Present CEO:** ByteDance Management
- **Tagline:** "Your Songs, Your Way"
- **Purpose:** Music streaming app with interactive features like lyrics sharing and social engagement.

These platforms cater to various multimedia needs, including **video streaming, live broadcasting, music streaming, podcasting, and user-generated content**, making them integral parts of digital entertainment.

8. Hypertext

Definition:

Hypertext is a way of organizing and linking digital information, allowing users to jump from one piece of text to another by clicking on links. It is the foundation of the **World Wide Web (WWW)**, where webpages are connected through hyperlinks.

Origin:

- The concept of **Hypertext** was first introduced by **Ted Nelson** in **1965**.
- The term "Hypertext" means "non-linear text," where users can navigate freely instead of reading in a fixed sequence.

History:

- **1945:** Vannevar Bush proposed a system called **Memex**, an early idea of linked information.

- **1965:** Ted Nelson coined the term **Hypertext**.
- **1980s:** Tim Berners-Lee developed **HTML (Hypertext Markup Language)**, which became the base of the **World Wide Web**.
- **1990s-Present:** The internet and websites expanded, using hypertext for linking web pages.

Key Features of Hypertext:

1. **Non-Linear Navigation:** Users can move between linked documents instead of following a fixed order.
2. **Clickable Links (Hyperlinks):** Words, images, or buttons can connect to other pages or files.
3. **Interactive Experience:** Users control what they read and explore.
4. **Foundation of the Web:** Websites use **HTML (Hypertext Markup Language)** to create linked content.
5. **Multimedia Integration:** Hypertext can include text, images, videos, and sounds.

Examples of Hypertext:

- **Websites:** Clicking links on Wikipedia to move between topics.
- **E-books:** Jumping to different chapters using hyperlinks.
- **Online Courses:** Clicking on lessons or references for more details.

9. Interactivity

Definition:

Interactivity means a two-way communication where users can actively engage with digital content, systems, or people. It allows users to **control, respond, or influence** what happens on a website, app, or device.

Origin:

- The concept of **Interactivity** started in the **1950s-60s** with early computer systems that allowed user input.
- The term became more common in the **1980s-90s** with the rise of **multimedia, video games, and the internet**.

History:

- **1950s-60s:** Early computers used basic interactivity (input/output commands).

- **1970s-80s:** Video games and interactive software became popular.
- **1990s:** The internet allowed real-time interactions (chat rooms, websites).
- **2000s-Present:** Social media, touchscreens, and AI-powered chatbots enhanced interactivity.

Key Features of Interactivity:

1. **User Control:** Users can choose actions (clicking, typing, touching, etc.).
2. **Real-Time Response:** Instant feedback (e.g., live chats, online quizzes).
3. **Engagement:** Interactive content keeps users interested.
4. **Two-Way Communication:** Unlike traditional media (TV, newspapers), users can respond or participate.
5. **Variety of Forms:** Found in websites, video games, social media, and virtual reality (VR).

Examples of Interactivity:

- **Websites:** Clicking on links, filling forms, using online maps.
- **Social Media:** Commenting, liking, and sharing posts.
- **Video Games:** Players control characters and actions.
- **Educational Apps:** Online quizzes and virtual classrooms.

10. Dropbox

Definition:

Dropbox is a cloud storage service that allows users to save, access, and share files online. It helps people store documents, photos, and videos securely and access them from any device with an internet connection.

Origin:

- **Dropbox** was founded in **2007** by **Drew Houston and Arash Ferdowsi**.
- The idea came when Drew Houston kept forgetting his USB drive and wanted an easier way to access files from anywhere.

History:

- **2007:** Dropbox was launched as a startup.
- **2008:** The first version was released to the public.

- **2010s:** Dropbox became one of the most popular cloud storage services.
- **Present:** It offers advanced features like automatic backup, team collaboration, and integration with apps like Google Docs and Microsoft Office.

Key Features of Dropbox:

1. **Cloud Storage:** Saves files online, reducing the need for USB drives.
2. **Easy Access:** Files can be accessed from computers, phones, and tablets.
3. **File Sharing:** Users can share documents with others through links.
4. **Automatic Syncing:** Changes in files are updated across all devices.
5. **Security & Backup:** Provides encryption and file recovery options.

Examples of Dropbox Use:

- **Personal:** Storing family photos and important documents.
- **Business:** Sharing work files with team members.
- **Education:** Students and teachers can store and access study materials.

11. Timeline

Definition:

A **timeline** is a way of displaying events in chronological order (from past to present). It helps people understand history, project plans, or important dates by organizing information in a visual format.

Origin:

- The concept of a timeline has been used for centuries in history, maps, and records.
- The modern **graphical timeline** was introduced in the **18th and 19th centuries** for historical and scientific studies.

History:

- **Ancient Times:** People recorded events on stone, papyrus, and scrolls.
- **18th-19th Century:** Scholars and historians began using visual timelines to explain events.
- **20th Century:** Timelines were widely used in books, education, and business.

- **21st Century:** Digital timelines became common in software, social media (Facebook Timeline), and project management.

Key Features of a Timeline:

1. **Chronological Order:** Events are arranged by date and time.
2. **Visual Representation:** Uses lines, dates, and images to display events.
3. **Helps in Understanding History:** Shows historical events, company growth, or personal achievements.
4. **Used in Different Fields:** Found in education, business, project management, and social media.
5. **Interactive & Digital Formats:** Modern timelines can be created using software like Microsoft Project, Trello, and Facebook Timeline.

Examples of Timelines:

- **Historical Timeline:** Important events in world history.
- **Project Timeline:** Steps to complete a task or business goal.
- **Personal Timeline:** Life events, such as school, jobs, and achievements.

12. Wordle

Definition:

Wordle is an online word puzzle game where players have to guess a **five-letter word** in **six attempts**. After each guess, the game gives color-coded hints to help players find the correct word.

Origin:

- **Wordle** was created in **2021** by **Josh Wardle**, a software engineer from the United Kingdom.
- He made the game for fun and later shared it with the public.

History:

- **October 2021:** Wordle was launched online.
- **January 2022:** The New York Times bought Wordle due to its huge popularity.
- **Present:** The game remains free and widely played around the world.

Key Features of Wordle:

1. **Simple Gameplay:** Guess a five-letter word in six tries.
2. **Color Hints:**
 - ☐ **Green** = Correct letter in the right spot.
 - ☐ **Yellow** = Correct letter in the wrong spot.
 - ☐ **Gray** = Letter is not in the word.
3. **Daily Puzzle:** Only one new word is available each day.
4. **No Ads & Free to Play:** The game is free and has no distractions.
5. **Shareable Results:** Players can share their scores without spoilers.

Examples of Wordle-Like Games:

- **Quordle:** Solve four words at once.
- **Heardle:** Guess a song based on short clips.
- **Nerdle:** A math-based guessing game.

13. Word Cloud

Definition:

A **Word Cloud** is a visual representation of text data where frequently used words appear larger and bolder. It is used to highlight key terms in articles, speeches, or surveys.

Origin:

- The concept of word clouds started in the **1990s** with **text analysis** in data visualization.
- It became popular in the **2000s** with the rise of digital tools and big data analysis.

History:

- **1990s:** Early use in text mining and information visualization.
- **2000s:** Online tools like "Tag Clouds" became popular in websites and blogs.
- **2010s-Present:** Used widely in marketing, research, and social media analysis.

Key Features of Word Cloud:

1. **Text Visualization:** Displays words in different sizes based on frequency.
2. **Easy to Read:** Highlights important terms quickly.
3. **Customizable:** Users can change colors, fonts, and layouts.
4. **Useful for Analysis:** Helps in summarizing large amounts of text.

5. **Used in Many Fields:** Found in education, business, social media, and research.

Examples of Word Cloud Uses:

- **Education:** Summarizing key terms from books.
- **Marketing:** Analyzing customer feedback.
- **Social Media:** Tracking trending topics.
- **Business Reports:** Identifying important keywords in reviews.

14. Open Source Software

Definition:

Open Source Software (OSS) is software that is freely available for anyone to use, modify, and share. Its source code is open to the public, allowing developers to improve and customize it.

Origin:

- The idea of **sharing software freely** started in the **1950s-60s** when early programmers collaborated.
- The term "**Open Source**" was officially introduced in **1998** by a group of software developers.

History:

- **1950s-60s:** Early computer programs were shared among researchers.
- **1980s:** Richard Stallman launched the **Free Software Movement** and created the **GNU Project**.
- **1991:** **Linux**, a famous open-source operating system, was developed by Linus Torvalds.
- **2000s-Present:** Open source became widely used in business, education, and government.

Key Features of Open Source Software:

1. **Free to Use:** Anyone can download and use it.
2. **Open Source Code:** Users can view, edit, and improve the software.
3. **Community Support:** Developed and maintained by a global community of programmers.
4. **Transparency:** No hidden codes or secret tracking by companies.
5. **Cost-Effective:** Saves money compared to paid software.

Examples of Open Source Software:

- **Operating Systems:** Linux, Ubuntu.
- **Web Browsers:** Mozilla Firefox.
- **Office Software:** LibreOffice, OpenOffice.
- **Programming Tools:** Python, PHP, MySQL.

15. Phishing

Definition:

Phishing is a type of cybercrime where attackers trick people into sharing personal information like passwords, bank details, or credit card numbers. They usually do this by sending fake emails, messages, or websites that look real.

Origin:

- The term "**Phishing**" was first used in the **1990s** when hackers targeted users of **AOL (America Online)** with fake login pages.
- It comes from the word "fishing," as attackers "bait" victims to steal their data.

History:

- **1990s:** First phishing attacks targeted AOL users through fake login pages.
- **2000s:** Phishing expanded to emails pretending to be from banks or companies.
- **2010s-Present:** Advanced phishing attacks use fake websites, SMS, and social media messages.

Key Features of Phishing:

1. **Fake Emails & Messages:** Looks like real emails from banks, companies, or friends.
2. **Deceptive Links:** Links take users to fake websites that steal passwords.
3. **Urgency & Fear Tactics:** Messages create panic (e.g., "Your account will be blocked!").
4. **Mass Targeting:** Sent to thousands of users at once through email or SMS.
5. **Difficult to Detect:** Some phishing emails bypass spam filters.

Examples of Phishing Attacks:

- **Email Phishing:** Fake emails from banks or companies asking for login details.
- **Spear Phishing:** Targeted attacks on specific people or businesses.

- **Smishing (SMS Phishing):** Fake text messages with scam links.
- **Vishing (Voice Phishing):** Phone calls pretending to be from banks or government agencies.

- **Worm:** Spreads automatically without needing a host file.
- **Trojan Horse:** Disguises itself as useful software but is harmful.
- **Spyware:** Secretly collects user information.
- **Ransomware:** Locks files and demands money to unlock them.

16. Malware

Definition:

Malware (Malicious Software) is a harmful program designed to damage, steal data, or gain unauthorized access to a computer or network. It includes viruses, worms, trojans, spyware, and ransomware.

Origin:

- The concept of malware started in the **1970s** when early computer viruses were created as experiments.
- The term "**malware**" became widely used in the **1990s** as cybercriminals started using it for attacks.

History:

- **1971:** The first virus, **Creeper**, was created as an experiment.
- **1980s:** The first PC viruses like **Brain (1986)** and **Morris Worm (1988)** spread widely.
- **1990s:** Malware became a tool for cybercrime, affecting personal and business computers.
- **2000s-Present:** Advanced malware like ransomware, spyware, and trojans target individuals, companies, and governments.

Key Features of Malware:

1. **Harmful Effects:** Can delete files, steal data, or slow down computers.
2. **Different Types:** Includes viruses, worms, trojans, spyware, and ransomware.
3. **Spreads in Many Ways:** Through email attachments, fake software, USB drives, and malicious websites.
4. **Hard to Detect:** Some malware hides in files or software updates.
5. **Can Be Prevented:** Using antivirus software, strong passwords, and avoiding suspicious links.

Examples of Malware Types:

- **Virus:** Attaches to files and spreads when opened.

17. Identity

Definition:

Identity in the digital world refers to a person's online information, such as usernames, passwords, emails, and social media profiles. **Identity security issues** arise when hackers steal or misuse this information for fraud, hacking, or identity theft.

Origin:

- The concept of **digital identity** started in the **1990s** with the rise of the internet.
- As online banking, social media, and e-commerce grew, identity security became a major concern.

Key Security Issues in Digital Identity:

1. **Identity Theft:** Hackers steal personal information for fraud.
2. **Phishing Attacks:** Fake emails and websites trick people into giving passwords.
3. **Data Breaches:** Companies storing personal data get hacked, exposing user details.
4. **Weak Passwords:** Simple passwords make accounts easy to hack.
5. **Social Media Risks:** Personal details shared online can be misused.
6. **Biometric Data Theft:** Hackers can steal fingerprints and facial recognition data.

Examples of Digital Identity Risks:

- **Bank Fraud:** Stolen login details used to withdraw money.
- **Fake Social Media Accounts:** Criminals create fake profiles to scam people.
- **Hacked Emails:** Cybercriminals access emails to steal sensitive information.

Ways to Protect Digital Identity:

- ☐ Use **strong passwords** and enable **multi-factor authentication (MFA)**.
- ☐ Avoid clicking on **suspicious emails or links**.

- ☐ Keep software and antivirus updated.
- ☐ Be careful when sharing personal information online.

18. Thefts

Definition:

Theft in digital technology refers to cybercrimes where criminals steal personal, financial, or business data from computers, smartphones, or online accounts. This can include hacking, identity theft, phishing, and financial fraud.

Origin:

- Digital theft started in the **1990s** when the internet became popular and people began storing personal information online.
- Cybercriminals started using viruses, hacking tools, and phishing scams to steal data.

History:

- **1990s:** Early hacking cases and online fraud began.
- **2000s:** Cybercriminals started using phishing emails, spyware, and credit card fraud.
- **2010s-Present:** Advanced threats like ransomware, biometric data theft, and large-scale data breaches emerged.

Key Security Issues in Digital Theft:

1. **Identity Theft:** Criminals steal personal details like passwords and bank accounts.
2. **Phishing Attacks:** Fake emails and websites trick users into revealing sensitive information.
3. **Financial Fraud:** Hackers steal money through online banking scams.
4. **Data Breaches:** Companies get hacked, exposing customer information.
5. **Ransomware Attacks:** Hackers lock files and demand money to unlock them.
6. **Social Media Scams:** Criminals use fake profiles to trick people.

Examples of Digital Theft Risks:

- **Credit Card Fraud:** Stolen card details used for illegal transactions.
- **Hacked Social Media Accounts:** Personal photos and messages misused.
- **Fake Shopping Websites:** Scam sites steal money and personal details.

Ways to Prevent Digital Theft:

- ☐ Use **strong passwords** and enable **multi-factor authentication (MFA)**.
- ☐ Avoid **suspicious emails, messages, or links**.
- ☐ Keep **antivirus and software updated**.
- ☐ Do not share **personal details on public websites**.

19. Blogging & Vlogging

Definition:

- **Blogging** is writing and sharing content on a website, called a **blog**. It includes articles, reviews, and personal experiences.
- **Vlogging (Video Blogging)** is creating and sharing video content on platforms like YouTube, instead of writing.

Process of Blogging and Vlogging

Blogging Process

1. **Choose a Niche** – Select a topic of interest (e.g., travel, tech, health).
2. **Set Up a Blog** – Use platforms like WordPress, Blogger, or Medium.
3. **Create Content** – Write engaging, SEO-friendly articles with images and links.
4. **Publish & Promote** – Share on social media, forums, and email newsletters.
5. **Monetize** – Earn through ads, affiliate marketing, or sponsored posts.
6. **Engage & Update** – Respond to comments and update old content.

Vlogging Process

1. **Choose a Topic** – Decide on a subject (e.g., lifestyle, gaming, tutorials).
2. **Plan & Script** – Outline key points to make the video engaging.
3. **Record Video** – Use a good camera and microphone for quality.
4. **Edit & Enhance** – Use software like Adobe Premiere Pro or CapCut.
5. **Upload & Optimize** – Post on YouTube or other platforms with proper SEO tags.
6. **Promote & Engage** – Share on social media and interact with viewers.
7. **Monetize** – Earn through ads, sponsorships, and memberships.

Origin:

- **Blogging** started in the **1990s** as online diaries and personal journals.
- **Vlogging** began in the **2000s** when video-sharing platforms like **YouTube (2005)** became popular.

History:

- **1994:** The first blog was created by Justin Hall, called "Links.net."
- **1999:** The term "**blog**" was introduced. Blogging platforms like Blogger and WordPress were launched.
- **2005:** YouTube was launched, making vlogging popular.
- **2010s-Present:** Social media platforms like Instagram, TikTok, and Facebook made vlogging even more popular.

Key Features of Blogging:

1. **Text-Based Content:** Articles, guides, and reviews.
2. **SEO Optimization:** Helps websites rank higher on Google.
3. **Monetization:** Bloggers earn through ads, sponsorships, and affiliate marketing.
4. **Personal or Business Use:** Can be for hobbies, news, or company promotion.
5. **Easy to Create:** Platforms like WordPress and Blogger make it simple.

Key Features of Vlogging:

1. **Video-Based Content:** Visual storytelling through recorded videos.
2. **Platforms:** YouTube, TikTok, Instagram Reels.
3. **Engagement:** Viewers like, comment, and share videos.
4. **Monetization:** Vloggers earn from ads, sponsorships, and brand deals.
5. **Equipment Needed:** Camera, microphone, and editing software.

Examples of Blogging & Vlogging:

- **Blogging:** News websites, travel blogs, food blogs.
- **Vlogging:** YouTube travel vlogs, makeup tutorials, gaming videos.

20. HTML

Definition:

HTML (HyperText Markup Language) is the standard language used to create and design web pages. It structures web content using elements like headings, paragraphs, images, and links.

Origin:

- HTML was developed by **Tim Berners-Lee** in **1991** to create and share documents on the World Wide Web (WWW).

History:

- **1991:** The first version of HTML was introduced.
- **1995:** HTML 2.0 was released with improved features.
- **1999:** HTML 4.01 added better multimedia and scripting support.
- **2014:** HTML5 was introduced with modern features like video, audio, and animations.

Key Features of HTML:

1. **Structure of Web Pages:** Defines headings, text, images, and links.
2. **Uses Tags & Elements:** Example:
`<h1>Heading</h1>`,
`<p>Paragraph</p>`.
3. **Hyperlinks & Navigation:** Connects different web pages using `<a>` tags.
4. **Multimedia Support:** Supports images, audio, and videos.
5. **Works with CSS & JavaScript:** Enhances design and interactivity.

Examples of HTML Usage:

- Creating web pages.
- Displaying images and videos.
- Adding links and buttons.
- Structuring content for blogs and online stores.

21. FTP

Definition:

FTP (File Transfer Protocol) is a network protocol used to transfer files between a computer and a server over the internet. It helps upload, download, and manage files on websites and online storage.

Origin:

- **FTP** was developed in **1971** by **Abhay Bhushan** at MIT (Massachusetts Institute of Technology).
- It was created to allow file sharing between computers connected to the early internet (ARPANET).

History:

- **1971:** First version of FTP was introduced.
- **1980s:** Improved versions added more security and functionality.
- **1990s-Present:** FTP became widely used for website management and cloud storage.
- **Today:** Secure versions like **SFTP (Secure FTP)** and **FTPS (FTP Secure)** are used for better security.

Key Features of FTP:

1. **File Transfer:** Upload and download files between computers and servers.
2. **Remote Access:** Users can manage files stored on a remote server.
3. **Multiple Connections:** Supports large file transfers and multiple users.
4. **Authentication:** Requires a username and password for secure access.
5. **Supports Different Modes:**
 - **Active Mode:** Server connects back to the client.
 - **Passive Mode:** Client connects to the server (better for firewalls).

Examples of FTP Usage:

- Uploading website files to a server.
- Downloading software or documents from online servers.
- Transferring large files between companies or remote teams.
- Managing cloud storage or backups.

22. Hackers

Definition:

A **hacker** is a person who uses computer skills to access or modify systems, networks, or software, sometimes with permission and sometimes illegally. Hackers can be ethical (helpful) or malicious (harmful).

Origin:

- The term "**hacker**" first appeared in the **1960s** at **MIT (Massachusetts Institute of Technology)**, where students explored and modified computer programs to improve them.

Key Types of Hackers:

1. **White Hat Hackers (Ethical Hackers):** Work legally to find and fix security issues.
2. **Black Hat Hackers (Criminal Hackers):** Break into systems for illegal purposes like theft and damage.
3. **Gray Hat Hackers:** Sometimes break rules but do not harm users or systems.
4. **Script Kiddies:** Use pre-made hacking tools without deep knowledge.
5. **Hactivists:** Hack for political or social causes.

Key Features of Hackers:

- **Expert in Coding & Security:** Skilled in programming and networks.
- **Penetration Testing:** Find and fix security loopholes.
- **Use of Hacking Tools:** Software like Kali Linux, Metasploit, etc.
- **Impact on Society:** Can be good (protecting data) or bad (stealing information).

Examples of Hacking Activities:

- **Cyber Attacks:** Hacking government or company data.
- **Ethical Hacking:** Security experts testing networks for safety.
- **Phishing & Malware Attacks:** Stealing personal information online.

23. Web 2.0

Definition:

Web 2.0 refers to the second generation of the internet, where websites became **interactive, user-driven, and social**. Unlike Web 1.0 (which was mostly static), Web 2.0 allows users to create, share, and collaborate online.

Origin:

- The term "**Web 2.0**" was first used by **Darcy DiNucci** in **1999**.
- It became popular after the **Web 2.0 Conference in 2004**, led by **Tim O'Reilly**.

History:

- **1990s (Web 1.0):** Websites were static, and users could only read content.
- **Early 2000s (Web 2.0):** Social media, blogs, and user-generated content became common.
- **2010s-Present:** Mobile apps, cloud computing, and AI-powered platforms dominate Web 2.0.

Key Features of Web 2.0:

1. **User Interaction:** Users can like, comment, and share content.
2. **Social Media & Networking:** Platforms like Facebook, Twitter, and YouTube allow global connections.
3. **User-Generated Content:** Blogs, videos, and reviews created by users.
4. **Cloud Computing:** Data is stored and accessed online instead of local computers.
5. **Rich Media & Streaming:** Support for videos, audio, and animations (e.g., YouTube, Spotify).

Examples of Web 2.0:

- **Social Media:** Facebook, Instagram, Twitter.
- **Video Platforms:** YouTube, TikTok.
- **Collaborative Tools:** Google Docs, Wikipedia.
- **E-Commerce:** Amazon, eBay (with user reviews and feedback).

Web (World Wide Web - WWW)

Definition:

The **Web (World Wide Web)** is a system of connected websites and web pages that can be accessed through the internet using a web browser like Chrome or Firefox.

Origin:

- **Invented by Tim Berners-Lee in 1989** while working at CERN.

History:

- **1991:** The first website was created.
- **1990s:** Websites were mostly **static (Web 1.0)** with only text and images.
- **2000s-Present:** The internet became **interactive and user-driven (Web 2.0 and beyond)**.

Key Features of the Web:

1. **Hyperlinked Information:** Websites are connected through links.
2. **Accessible via Browsers:** Can be viewed using Chrome, Firefox, etc.

3. **Contains Text, Images, and Videos:** Supports multimedia content.
4. **Global Connectivity:** Used for communication, business, and education.

24. Generation X (Gen X)

Definition:

Generation X (Gen X) refers to people born between **1965 and 1980**. They grew up during a time of rapid technological change and are known for being independent, hardworking, and adaptable.

Origin:

- The term "**Generation X**" was first used by photographer **Robert Capa** in the **1950s**.
- It became widely popular after **Douglas Coupland's 1991 book "Generation X: Tales for an Accelerated Culture."**

History:

- **1965-1980:** Gen X was born during a period of economic and social changes.
- **1980s-1990s:** They entered adulthood, witnessing the rise of computers, cable TV, and early internet.
- **2000s-Present:** Now in leadership roles, balancing traditional work ethics with digital skills.

Key Features of Generation X:

1. **Independent & Self-Reliant:** Many grew up with working parents and learned to take care of themselves.
2. **Tech-Savvy:** Experienced both pre-digital and digital worlds (landlines to mobile phones, typewriters to computers).
3. **Work-Life Balance:** Value both career success and personal life.
4. **Highly Educated:** Many pursued higher education and professional careers.
5. **Cultural Influence:** Witnessed major global events like the Cold War, fall of the Berlin Wall, and rise of MTV culture.

Examples of Generation X Influences:

- Adapted to new technology (computers, internet, social media).
- Shaped workplace culture with teamwork and problem-solving skills.

- Influenced music, movies, and fashion trends of the 1980s-90s.

25. Generation Z (Gen Z) (Zoomers)

Definition:

Generation Z (Gen Z), also called **Zoomers**, refers to people born between **1997 and 2012** (approximate range). They are the first generation to grow up fully in the digital age, surrounded by the internet, smartphones, and social media.

Origin:

- The term "**Generation Z**" was first used in the **early 2000s** to describe the generation after Millennials.
- The nickname "**Zoomers**" became popular as a playful version of "Boomers" (Baby Boomers), referring to their fast-paced, tech-driven lives.

History:

- **1997-2012:** Gen Z was born during the rise of the internet and digital technology.
- **2000s-2010s:** Grew up using smartphones, YouTube, Instagram, and later TikTok.
- **2020s-Present:** Entering the workforce, shaping trends in social media, gaming, and online activism.

Key Features of Generation Z (Zoomers):

1. **Tech-Savvy:** Grew up with the internet, smartphones, and digital tools.
2. **Social Media Experts:** Active on TikTok, Instagram, Snapchat, and YouTube.
3. **Short Attention Span:** Prefer fast and engaging content like short videos and memes.
4. **Diverse & Inclusive:** Support social justice, equality, and climate activism.
5. **Entrepreneurial Mindset:** Interested in freelancing, startups, and digital careers.

Examples of Generation Z (Zoomers) Influences:

- Popularized **TikTok, YouTube Shorts, and Instagram Reels**.
- Advocate for **climate change, gender equality, and mental health awareness**.
- Prefer **online shopping, digital banking, and remote work** over traditional methods.

26. Generation Alpha

Definition:

Generation Alpha refers to people born between **2013 and 2025** (approximate range). They are the first generation to be fully born in the digital age, growing up with smartphones, artificial intelligence (AI), and smart technology from birth.

Origin:

- The term "**Generation Alpha**" was introduced by **Mark McCrindle**, an Australian researcher, in **2008**.
- It was named "Alpha" because it is the first generation of the 21st century, following **Generation Z**.

History:

- **2013-Present:** Generation Alpha is growing up in a world dominated by digital learning, social media, and automation.
- **2020s:** The COVID-19 pandemic influenced their early education with online learning and virtual classrooms.

Key Features of Generation Alpha:

1. **Born in the Digital Age:** Raised with AI, smartphones, and smart devices.
2. **Tech-Dependent:** Prefer digital learning, online games, and virtual reality.
3. **Shorter Attention Span:** Engage more with short videos, interactive content, and voice assistants.
4. **Highly Educated & Informed:** Learn through digital platforms like YouTube and educational apps.
5. **Global & Socially Aware:** Exposed to worldwide issues like climate change and social justice from a young age.

Examples of Generation Alpha Influences:

- Use of **AI-powered toys, tablets, and voice assistants (Alexa, Siri)**.
- Prefer **interactive learning apps and digital classrooms**.
- Early exposure to **social media, influencers, and virtual reality (VR)**.

27. Sexting

Definition:

Sexting is the act of sending or receiving **sexual messages, photos, or videos** using mobile phones, social media, or messaging apps. It can be done between partners but may also lead to risks like privacy violations and cybercrime.

Origin:

- The term "**Sexting**" is a combination of "**Sex**" and "**Texting**".
- It became popular in the **early 2000s** with the rise of mobile phones and instant messaging.

History:

- **1990s:** Early forms of sexting happened through emails and SMS.
- **2000s:** With smartphones and social media, sharing private images and videos became common.
- **2010s-Present:** Apps like Snapchat, WhatsApp, and Instagram increased sexting, leading to privacy concerns and cybercrimes.

Key Features of Sexting:

1. **Digital Sharing:** Involves sending texts, photos, or videos with sexual content.
2. **Private & Risky:** Can be shared between trusted people but also misused.
3. **Legal & Ethical Issues:** In some countries, sexting among minors is illegal.
4. **Cybersecurity Risks:** Hackers or scammers may leak or misuse private content.
5. **Emotional & Social Impact:** Can lead to embarrassment, blackmail, or cyberbullying.

Examples of Sexting Risks:

- **Leaked Photos:** Private images can be shared without consent.
- **Cyberbullying & Harassment:** Sexting content can be used for blackmail.
- **Legal Issues:** Some laws consider sharing explicit content a criminal offense, especially among minors.

Safety Tips:

- ☐ Never share private images with untrusted people.
- ☐ Use strong privacy settings on social media.
- ☐ Report and block users who demand explicit content.

28. Citizen Journalism

Definition:

Citizen Journalism is when ordinary people (not professional journalists) report news and share information using mobile phones, social media, blogs, or websites. It allows the public to participate in news reporting.

Origin:

- The idea of **citizen journalism** has existed for centuries, as people have always shared news within their communities.
- The modern concept became popular in the **2000s** with the rise of the internet, social media, and smartphones.

History:

- **17th-18th Century:** Citizens wrote and distributed pamphlets about political and social issues.
- **1990s:** The internet allowed people to post news online through blogs and forums.
- **2000s-Present:** Social media platforms like Twitter, Facebook, and YouTube helped spread news instantly. Major events (like protests, disasters, and political movements) were reported first by ordinary people.

Key Features of Citizen Journalism:

1. **Non-Professional Reporting:** Anyone with a smartphone or internet can report news.
2. **Real-Time Updates:** News spreads quickly through social media.
3. **Unfiltered & Raw Content:** Reports may lack professional editing but provide direct coverage.
4. **Wide Reach:** Can reach millions of people without traditional media.
5. **Challenges:** Risks of fake news, misinformation, and lack of journalistic ethics.

Examples of Citizen Journalism:

- A person live-streaming a protest on Facebook.
- A local resident sharing photos of a natural disaster on Twitter.
- A blogger exposing corruption in their community.

29. Computer-Assisted Reporting (CAR)

Definition:

Computer-Assisted Reporting (CAR) is the use of computers, software, and data analysis tools to gather, process, and report news. It helps journalists find patterns, verify facts, and present data in an easy-to-understand way.

Origin:

- The concept of **CAR** started in the **1950s-60s** when journalists used computers to analyze large datasets.
- It became more common in the **1990s** with the rise of the internet and digital databases.

Key Features of Computer-Assisted Reporting:

1. **Data Collection:** Journalists use databases, spreadsheets, and online sources to gather information.
2. **Data Analysis:** Helps find patterns, trends, and hidden facts.
3. **Fact-Checking & Verification:** Ensures news accuracy using technology.
4. **Visualization & Presentation:** Uses graphs, charts, and interactive maps to explain complex stories.
5. **Faster & More Efficient:** Computers speed up investigative journalism.

Examples of CAR in Journalism:

- **Election Reporting:** Analyzing voting results using computer programs.
- **Crime Investigations:** Using databases to track crime rates and trends.
- **Financial Reporting:** Examining bank records and business reports for corruption.
- **Health Journalism:** Tracking disease outbreaks with government data.

30. Cyber Journalism

Definition:

Cyber Journalism refers to news reporting and publishing using the internet and digital platforms. It includes online newspapers, news websites, blogs, and social media-based news reporting.

Origin:

- The concept of **Cyber Journalism** started in the **1990s** with the rise of the internet.
- The first online newspaper was **The Chicago Tribune** in **1992**.

History:

- **1990s:** Traditional newspapers launched websites for online news.
- **2000s:** Social media (Facebook, Twitter, YouTube) became platforms for news sharing.
- **2010s-Present:** Mobile apps, live streaming, and AI-powered news recommendations made cyber journalism faster and more interactive.

Key Features of Cyber Journalism:

1. **Instant News Updates:** News is reported in real-time on websites and social media.
2. **Global Reach:** Information is available to people worldwide.
3. **Multimedia Content:** Uses text, images, videos, and live streaming.
4. **Interactivity:** Readers can comment, share, and engage with news.
5. **Cost-Effective:** No need for printing or physical distribution.

Examples of Cyber Journalism:

- **Online News Portals:** BBC News, CNN, The Hindu Online.
- **Social Media News:** Twitter news updates, Facebook Live reports.
- **Citizen Journalism:** People reporting news through blogs and YouTube.
- **Podcasts & Digital Magazines:** Audio and digital news analysis.

31. Convergence Media

Definition:

Convergence Media refers to the blending of different forms of media (print, TV, radio, internet, and social media) into a single digital platform. It allows users to access news, entertainment, and information anytime and anywhere using various devices.

Origin:

- The concept started in the **1990s** with the rise of digital technology and the internet.
- The term became popular with advancements in **multimedia, mobile communication, and the internet**.

History:

- **1990s:** Traditional media (newspapers, TV, radio) started using websites.
- **2000s:** Social media, mobile apps, and video streaming became common.
- **2010s-Present:** AI, virtual reality (VR), and smart devices improved media accessibility and interaction.

Key Features of Convergence Media:

1. **Integration of Multiple Media:** Combines text, audio, video, and graphics.
2. **Interactive & Engaging:** Users can comment, share, and participate in discussions.
3. **Accessible on Different Devices:** Available on smartphones, tablets, computers, and smart TVs.
4. **Real-Time Updates:** News and content are updated instantly.
5. **User-Generated Content:** People contribute content through blogs, vlogs, and social media.

Examples of Convergence Media:

- **Online News Portals:** Websites like BBC, CNN, and The Hindu combining articles, videos, and live updates.
- **Social Media News:** Facebook, Twitter, and YouTube spreading news instantly.
- **Streaming Platforms:** Netflix, YouTube, and Spotify offering video and audio content.
- **E-Newspapers & Digital Magazines:** Traditional newspapers publishing online versions.

32. Disinformation

Definition:

Disinformation is false or misleading information that is **deliberately created and spread** to deceive people. It is different from misinformation, which is false information shared unknowingly.

Origin:

- The term "**disinformation**" comes from the Russian word "**dezinformatsiya**," used by the **Soviet Union in the 1920s** for propaganda.
- It became widely known during **World War II and the Cold War**, when false news was spread to mislead enemies.

History:

- **1920s-1940s:** Governments used disinformation as propaganda during wars.
- **Cold War (1945-1991):** Countries spread false news to manipulate public opinion.
- **2000s-Present:** Social media made disinformation spread faster, influencing politics, health, and social issues.

Key Features of Disinformation:

1. **Intentional Deception:** Created to mislead people for political, financial, or personal gain.
2. **Spreads Quickly:** Shared through social media, fake news websites, and messaging apps.
3. **Manipulates Public Opinion:** Used in politics, advertising, and conflicts.
4. **Difficult to Detect:** Often disguised as real news or facts.
5. **Can Cause Harm:** Leads to confusion, panic, or violence.

Examples of Disinformation:

- **Fake News:** False reports about elections, wars, or health crises.
- **Deepfake Videos:** Edited videos making people appear to say or do things they never did.
- **Hoaxes & Conspiracy Theories:** False stories spread to create fear or distrust.

Ways to Prevent Disinformation:

- ☐ Verify news from trusted sources.
- ☐ Cross-check facts before sharing.
- ☐ Use factchecking websites.
- ☐ Be cautious of emotionally charged headlines.

33. Misinformation

Definition:

Misinformation is false or incorrect information that is **shared without the intention to deceive**. People may spread it unknowingly, believing it to be true.

Origin:

- The concept of **misinformation** has existed for centuries, as false information has always spread through word of mouth, books, and newspapers.
- The term became widely used in the **20th century**, especially with the rise of mass media.

Key Features of Misinformation:

1. **False Information Without Intent:** The person sharing it believes it is true.
2. **Spreads Quickly:** Can go viral on social media and news platforms.
3. **Affects Public Understanding:** Can lead to confusion and wrong decisions.
4. **Difficult to Correct:** Once spread, it is hard to undo the damage.
5. **Common in Social Media & News:** Often appears in fake viral posts, rumors, and misinterpreted data.

Examples of Misinformation:

- **Health Rumors:** False cures for diseases spreading on WhatsApp.
- **Misquoted Statements:** Incorrect attributions to famous people.
- **Edited Images:** Photos changed to create a misleading story.

34. Fake News

Definition:

Fake News is false or misleading information presented as real news. It is created to mislead people, influence opinions, or make money through clicks and shares.

Origin:

- The concept of **fake news** has existed for centuries, with false stories appearing in newspapers and propaganda materials.
- The term became widely popular in the **2010s**, especially during political events and elections.

History:

- **Before the Internet:** Fake news was spread through newspapers, word of mouth, and propaganda.
- **20th Century:** Governments and media sometimes spread false stories for political or military purposes.
- **2000s-Present:** The internet and social media made fake news spread faster, influencing politics, health, and society.

Key Features of Fake News:

1. **False or Misleading Information:** Created to deceive people.

2. **Spreads Quickly:** Shared widely through social media, websites, and messaging apps.
3. **Clickbait Headlines:** Uses emotional or shocking titles to grab attention.
4. **No Reliable Sources:** Fake news articles lack verified facts or credible references.
5. **Can Cause Harm:** Leads to confusion, fear, or wrong decisions in politics, health, and daily life.

Examples of Fake News:

- **Political Lies:** False claims about election results.
- **Health Hoaxes:** Fake medical advice or cures for diseases.
- **Celebrity Death Hoaxes:** False reports about a famous person's death.

Ways to Identify & Stop Fake News:

- ☐ Check if the news is reported by trusted media sources.
- ☐ Verify information from multiple sources.
- ☐ Avoid sharing news with shocking or extreme headlines without checking facts.
- ☐ Use fact-checking websites like Snopes, BBC Reality Check, or AFP Fact Check.

35. Civil Society

Definition:

Civil Society refers to organizations, groups, and individuals that work for the welfare of society and are independent of the government. It includes NGOs, community groups, labor unions, and activists who promote social causes, human rights, and democracy.

Origin:

- The concept of civil society dates back to **Ancient Greece** when philosophers like **Aristotle** discussed the role of citizens in governance.
- The modern idea developed in the **17th and 18th centuries** with thinkers like **John Locke** and **Alexis de Tocqueville**, who emphasized people's role in shaping society.

History:

- **Ancient Times:** Greek and Roman societies had civic groups involved in governance.
- **Middle Ages:** Religious and trade organizations helped communities.
- **18th-19th Century:** Civil society grew with democracy and social movements (e.g., human rights, abolition of slavery).
- **20th Century-Present:** NGOs, environmental groups, and online activism play a major role in global issues.

Key Features of Civil Society:

1. **Independent from Government:** Works separately from political authorities.
2. **Promotes Public Interests:** Focuses on human rights, education, environment, and social justice.
3. **Non-Profit Organizations:** Many groups work without aiming for financial gain.
4. **Encourages Democracy & Participation:** Allows people to express their views and demand accountability.
5. **Diverse & Global:** Includes small local groups and large international NGOs like Amnesty International and Greenpeace.

Examples of Civil Society Organizations:

- **NGOs:** Red Cross, Oxfam, Greenpeace.
- **Human Rights Groups:** Amnesty International, Human Rights Watch.
- **Community Groups:** Local environmental or social service organizations.
- **Labor Unions & Student Movements:** Groups fighting for workers' and students' rights.

36. Open Source License

Definition:

An **Open Source License** is a legal agreement that allows people to freely **use, modify, and share** software. It ensures that the software's source code remains open to the public.

Origin:

- The idea of sharing software freely started in the **1950s-60s** when programmers collaborated without restrictions.
- The term "**Open Source**" was officially introduced in **1998** to promote free software with legal guidelines.

History:

- **1980s:** Richard Stallman launched the **Free Software Movement** and the **GNU General Public License (GPL)**.
- **1998:** The **Open Source Initiative (OSI)** was formed to define open source licensing rules.
- **2000s-Present:** Many open-source projects like **Linux, Mozilla Firefox, and Android** became popular.

Examples of Open Source Licenses:

- **GNU General Public License (GPL):** Requires modified versions to remain open-source.
- **MIT License:** Allows free use with fewer restrictions.
- **Apache License:** Permits modifications and distribution with some conditions.

Examples of Open Source Software:

- **Linux (Operating System)**
- **Mozilla Firefox (Web Browser)**
- **Android (Mobile OS)**
- **LibreOffice (Office Suite)**

37. TB & ZB

Definition:

- **TB (Terabyte):** A unit of digital storage equal to **1,024 Gigabytes (GB)**. It is commonly used to measure the storage capacity of hard drives, SSDs, and cloud storage.
- **ZB (Zettabyte):** A much larger unit of digital storage equal to **1,024 Exabytes (EB)** or **one trillion Terabytes (TB)**. It is used to measure global internet data and massive data centers.

Origin:

- The concept of digital storage measurement started in the **1950s** with Kilobytes (KB) and Megabytes (MB).
- As data usage grew, larger units like **GB, TB, PB (Petabyte), and ZB** were introduced.

History:

- **1950s-70s:** Data storage was measured in KB and MB.
- **1980s-90s:** GB became common in personal computers.
- **2000s-Present:** TB storage is now used in personal devices, while ZB is used for massive global data.

Key Features:

1. **Terabyte (TB):**
 - Common in **hard drives, SSDs, and cloud storage.**
 - Example: A **1 TB HDD** can store about **250,000 photos** or **500 hours of HD video.**
2. **Zettabyte (ZB):**
 - Used for **big data, cloud computing, and internet storage.**
 - Example: In **2021, the world's total digital data was estimated to be around 79 ZB.**

Examples of Usage:

- **TB Usage:** Laptops, external hard drives, game consoles (PlayStation, Xbox).
- **ZB Usage:** Cloud storage companies (Google Drive, Amazon Web Services), global internet data storage.

Data Storage Units: Storage Units in Order

1. **Bit (b)**
 - **Definition:** Smallest unit of data, representing either 0 or 1 in binary.
 - **Key Features:**
 - Used in digital communication and computing.
 - Forms the basis of all digital storage.

2. **Byte (B)**
 - **Definition:** 8 bits form 1 byte, used to store a single character.
 - **Key Features:**
 - Fundamental unit for text storage.
 - Used in coding and programming.
3. **Kilobyte (KB)**
 - **Definition:** 1 KB = 1,024 Bytes, used for small text files.
 - **Key Features:**
 - Common for documents and emails.
 - Example: A short email (without attachments) is around 2 KB.
4. **Megabyte (MB)**
 - **Definition:** 1 MB = 1,024 KB, used for images and MP3 songs.
 - **Key Features:**
 - Holds about 1 minute of MP3 music.
 - Example: A high-resolution photo is around 5 MB.
5. **Gigabyte (GB)**
 - **Definition:** 1 GB = 1,024 MB, used for videos and applications.
 - **Key Features:**
 - Common in smartphones and USB drives.
 - Example: A 2-hour HD movie is around 4 GB.
6. **Terabyte (TB)**
 - **Definition:** 1 TB = 1,024 GB, used for large storage like hard drives.
 - **Key Features:**
 - Standard unit for external hard disks.
 - Example: A 1 TB drive can store around 250,000 photos.
7. **Petabyte (PB)**
 - **Definition:** 1 PB = 1,024 TB, used in large servers and data centers.
 - **Key Features:**
 - Found in cloud storage and data warehouses.
 - Example: Facebook stores **over 100 PB** of user data.
8. **Exabyte (EB)**
 - **Definition:** 1 EB = 1,024 PB, used for massive data like internet traffic.
 - **Key Features:**

- Global digital data is measured in EB.
- Example: Total data on the internet in 2020 was around **40 EB**.

9. Zettabyte (ZB)

- **Definition:** 1 ZB = 1,024 EB, used to measure total worldwide data.
- **Key Features:**
 - Used in large-scale computing and AI research.
 - Example: Global internet data in 2025 is expected to exceed **175 ZB**.

10. Yottabyte (YB)

- **Definition:** 1 YB = 1,024 ZB, theoretical storage unit for future computing.
- **Key Features:**
 - Used in deep-space research and quantum computing.
 - Example: Theoretical limit of today's computing storage.

11. Brontobyte (BB)

- **Definition:** 1 BB = 1,024 YB, massive storage for AI and global computing.
- **Key Features:**
 - Hypothetical storage beyond existing technology.
 - Example: Could store all human knowledge and AI-generated data.

12. Geopbyte (GP)

- **Definition:** 1 GP = 1,024 BB, the largest hypothetical storage unit.
- **Key Features:**
 - Not yet used in any real-world applications.
 - Example: Future technologies may require GP storage.

Example for Understanding:

A **1 TB** hard drive can store about **500 hours of HD video**, while a **1 PB** storage can hold over **11,000 4K movies**.

38. I2P (Invisible Internet Project)

Definition:

I2P (Invisible Internet Project) is an anonymous network that allows users to communicate and browse securely by encrypting and routing internet traffic through multiple nodes. It is designed to protect privacy and prevent surveillance or tracking.

Origin & History:

- **Developed in Early 2000s** by a group of privacy-focused developers.
- **Official Release:** 2003 as an open-source project.
- Inspired by early anonymity tools like Tor but built for secure peer-to-peer communication.
- Used widely for anonymous messaging, file sharing, and browsing hidden services.

Key Features:

1. **Anonymity:** Hides users' IP addresses and encrypts all communication.
2. **Decentralized Network:** Uses a peer-to-peer (P2P) structure to prevent a single point of failure.
3. **End-to-End Encryption:** Ensures data security while transferring information.
4. **Garlic Routing:** Similar to Tor's Onion Routing but bundles multiple messages together for extra security.
5. **Hidden Services:** Users can host anonymous websites (called "eepsites") without revealing their real identity.
6. **Peer-to-Peer Communication:** Supports messaging, email, and file sharing securely.

Example:

If a journalist wants to communicate with sources without government surveillance, they can use I2P to send messages securely without revealing their identity.

39. P2P (Peer-to-Peer)

Definition:

P2P (Peer-to-Peer) is a network system where computers (peers) connect directly to share

resources, such as files or processing power, without relying on a central server.

Origin & History:

- **Concept Introduced:** In the 1960s with early computer networking.
- **Popularized in the 1990s** with **Napster (1999)**, a file-sharing service.
- **Evolved into various applications** like **BitTorrent (2001)**, **Blockchain (2008)**, and **Decentralized Finance (DeFi)**.
- Now used in file sharing, cryptocurrencies, gaming, and communications.

Key Features:

1. **Decentralized System:** No central authority, users (peers) connect directly.
2. **Efficient Resource Sharing:** Peers share bandwidth, storage, or processing power.
3. **Scalability:** Works well with a large number of users.
4. **Resilience:** No single point of failure; if one peer goes offline, the network remains active.
5. **Anonymity & Security:** Some P2P networks use encryption for secure transactions.

Example:

When people download a movie using **BitTorrent**, they are using P2P, where multiple users share parts of the file with each other instead of downloading from one central server.

40. Tor (The Onion Router)

Definition:

Tor is a free and open-source software that allows users to browse the internet anonymously by routing their traffic through multiple encrypted layers across different servers worldwide.

Origin & History:

- **Developed by the U.S. Naval Research Laboratory (1990s)** for secure government communications.

- **Released as Open Source (2002)** by Roger Dingledine and Nick Mathewson.
- **The Tor Project Founded (2006)** as a non-profit organization to maintain and improve Tor.
- **Widely Used Today** for privacy, journalism, and bypassing censorship.

Key Features:

1. **Anonymity:** Hides users' IP addresses by bouncing traffic through multiple nodes.
2. **Onion Routing:** Encrypts data in multiple layers, similar to peeling an onion.
3. **Access to .onion Sites:** Enables browsing of hidden websites that are not available on regular search engines.
4. **Decentralized Network:** No single point of control, ensuring better privacy.
5. **Bypasses Censorship:** Helps users in restricted regions access blocked websites securely.

Example:

A journalist in a country with strict internet laws can use **Tor** to safely communicate with sources and access uncensored information without revealing their identity.

41. ARPANET

Definition:

ARPANET (Advanced Research Projects Agency Network) was the first computer network to use packet-switching technology, laying the foundation for the modern internet.

Origin & History:

- **Developed by the U.S. Department of Defense (DARPA) in 1969** to create a secure and reliable communication system.
- **First message sent on October 29, 1969**, between UCLA and Stanford Research Institute.
- **Expanded in the 1970s**, connecting universities and research centers.
- **Introduced TCP/IP Protocol (1983)**, which later became the backbone of the internet.
- **ARPANET shut down in 1990**, as the internet took over.

Key Features:

1. **Packet-Switching Technology:** Broke data into small packets for efficient transfer.
2. **Decentralized Communication:** Designed to work even if some parts of the network failed.
3. **First Use of TCP/IP (1983):** Enabled standardized internet communication.
4. **Connected Research Institutions:** Allowed scientists to share data and collaborate.
5. **Foundation of the Internet:** Led to the development of modern networking technologies.

Example:

Before the internet, ARPANET allowed researchers from different universities to share data instantly instead of relying on postal mail or fax.

DARPA (Defense Advanced Research Projects Agency)

Definition:

DARPA is a research agency of the **U.S. Department of Defense**, responsible for developing advanced military technologies, including the **internet, GPS, and artificial intelligence (AI)**.

Origin & History:

- **Founded in 1958** by the U.S. government after the Soviet Union launched Sputnik (1957).
- **Created ARPANET in 1969**, which led to the internet.
- **Developed modern technologies** like stealth aircraft, voice recognition, and robotics.

Key Features:

1. **Defense & Security Focus:** Develops military and national security technologies.
2. **Innovation Leader:** Responsible for the internet, GPS, AI, and cybersecurity advancements.
3. **Collaboration with Private & Public Sectors:** Works with universities and tech companies.

4. **Rapid Development:** Invests in cutting-edge research projects.
5. **Global Impact:** Many DARPA projects influence civilian life, like the internet and GPS.

Example:

DARPA's invention of ARPANET in 1969 eventually led to today's **internet**, which billions of people use worldwide.

42. MOOC (Massive Open Online Course)

Definition:

MOOC (Massive Open Online Course) is an online learning platform that offers free or paid courses to a large number of students worldwide. It provides flexible and accessible education through video lectures, assignments, and interactive discussions.

Origin & History:

- **Concept Introduced in 2008** by George Siemens and Stephen Downes.
- **First Major MOOC (2011):** Stanford University's "Introduction to AI" course attracted 160,000 students.
- **Rise of Platforms (2012-Present):** Coursera, edX, Udacity, and FutureLearn became popular.
- **Used Globally Today** by universities, companies, and independent learners.

Key Features:

1. **Massive Enrollment:** Open to thousands or millions of students.
2. **Online Learning:** Courses are available anytime, anywhere.
3. **Interactive Content:** Includes video lectures, quizzes, and peer discussions.
4. **Certification:** Some platforms offer certificates for course completion.
5. **Affordable & Accessible:** Many courses are free or low-cost, making education available to all.

Example:

A student from India can take a **Harvard University course on edX** without traveling, gaining knowledge from top professors at no cost.

43. B2B (Business-to-Business)

Definition: B2B (Business-to-Business) is a type of commercial transaction where businesses sell products or services directly to other businesses instead of individual consumers.

Origin & History:

- **Concept existed for centuries** as businesses traded goods and services with each other.
- **Modern B2B grew in the 20th century** with industrialization and global trade.
- **Expansion in the 1990s** with the rise of the internet, making online B2B marketplaces popular.
- **Today, digital B2B platforms** like Alibaba, Amazon Business, and Salesforce dominate the market.

Key Features:

1. **Bulk Transactions:** Businesses buy in large quantities for resale or production.
2. **Long-Term Relationships:** Companies often have long-term contracts with suppliers.
3. **Higher Order Value:** B2B transactions involve more money compared to B2C (Business-to-Consumer).
4. **Negotiation & Customization:** Prices, terms, and products are often tailored to specific business needs.
5. **Technology-Driven:** Uses digital platforms, automation, and cloud solutions for efficiency.

Example:

A **car manufacturer (like Toyota)** buys steel from a **steel company** to make cars. This is a **B2B transaction** because both companies are businesses.

44. Evaluation and Growth of the Internet

Definition:

The Internet is a global network of interconnected computers that allows communication, data sharing, and access to information worldwide. It has evolved

from a military research project into a vital part of daily life, connecting billions of people.

Origin & History:

- **1960s:** The U.S. Department of Defense developed **ARPANET** as a secure communication system.
- **1970s:** Email was introduced, and networking protocols like **TCP/IP (1983)** standardized internet communication.
- **1990s:** The **World Wide Web (WWW)** was created by **Tim Berners-Lee (1989)**, leading to websites, search engines, and online businesses.
- **2000s-Present:** The rise of **social media, e-commerce, cloud computing, and 5G technology** expanded internet access globally.

Key Features & Growth Factors:

1. **Global Connectivity:** Billions of devices and users are connected worldwide.
2. **Fast Communication:** Enables instant messaging, video calls, and social media.
3. **Information Access:** Search engines like Google provide vast knowledge.
4. **E-Commerce Expansion:** Platforms like Amazon and Alibaba revolutionized shopping.
5. **Advancement in Technology:** The internet has integrated **AI, IoT, and blockchain** for smarter solutions.

A student in **India** can take an online class from **Harvard University** using the internet, showcasing its power in education and connectivity.

45. Types of Web

Definition:

The web (World Wide Web) is a system of interlinked documents and services accessible via the internet. Over time, different types of web have evolved based on usage, technology, and accessibility.

Types of Web:

1. **Surface Web (Visible Web)**
 - **Definition:** The part of the web that is indexed by search engines like Google and Bing.
 - **Origin:** Developed with the creation of the World Wide Web

(WWW) in **1989** by **Tim Berners-Lee**.

- **Features:**
 - Publicly accessible.
 - Includes news websites, social media, and e-commerce sites.
- **Example:** Wikipedia, YouTube, Facebook.

2. Deep Web (Hidden Web)

- **Definition:** The part of the web that is not indexed by search engines but still accessible with a login or direct URL.
- **Origin:** Evolved as private networks, databases, and academic resources grew online.
- **Features:**
 - Requires authentication (username and password).
 - Includes online banking, medical records, and government databases.
- **Example:** Email accounts, Netflix library, university portals.

3. Dark Web (Encrypted Web)

- **Definition:** A hidden part of the web that requires special software like **Tor (The Onion Router)** to access. It is often used for privacy and anonymity.
- **Origin:** Developed in the **1990s** by the **U.S. Naval Research Laboratory** for anonymous communication.
- **Features:**
 - Not indexed by search engines.
 - Used for both legal and illegal activities.
- **Example:** .onion websites, anonymous forums, cryptocurrency markets.

4. Web 1.0 (Static Web)

- **Definition:** The first generation of the web with **read-only** content and no user interaction.
- **Origin:** Introduced in the **1990s** with basic HTML pages.
- **Features:**
 - Static websites with text and images.
 - No user-generated content or interactivity.

- **Example:** Early news websites and company pages.

5. Web 2.0 (Dynamic & Social Web)

- **Definition:** A more interactive version of the web, allowing user-generated content and social networking.
- **Origin:** Popularized in the **2000s** with the rise of social media.
- **Features:**
 - Users can comment, share, and create content.
 - Blogging, social networking, and e-commerce grew rapidly.
- **Example:** Facebook, YouTube, Wikipedia.

6. Web 3.0 (Semantic & AI-Powered Web)

- **Definition:** The next stage of the web that uses AI, blockchain, and decentralized networks for a smarter and more personalized experience.
- **Origin:** Concept introduced in the **2010s**, still evolving.
- **Features:**
 - Artificial Intelligence (AI) for personalized searches.
 - Decentralization with blockchain for privacy and security.
- **Example:** Cryptocurrencies, smart contracts, AI-based recommendations.

Conclusion:

The web has evolved from a simple **static system (Web 1.0)** to an **interactive (Web 2.0)** and **AI-driven future (Web 3.0)** while also including hidden layers like the **Deep Web** and **Dark Web** for privacy and security.

Example:

A person using **Google (Surface Web)** to search for medical information, logging into a **hospital portal (Deep Web)** to view health records, and using **Tor to browse anonymously (Dark Web)** is engaging with different types of the web.